DSC 257R: Unsupervised learning
Homework 7
Markov random fields and energy-based models

1. Probability distribution P over binary variables $x_1, \ldots, x_6 \in \{0, 1\}$ has the following form:

$$P(x_1, \ldots, x_6) = \frac{1}{Z}\psi(x_1, x_2)\psi(x_2, x_3)\psi(x_3, x_4)\psi(x_4, x_5)\psi(x_5, x_6)\psi(x_6, x_1)$$

   where for $a, b \in \{0, 1\}$,

$$\psi(a, b) = \begin{cases} 1 & \text{if } a = b \\ 0.5 & \text{otherwise} \end{cases}$$

   (a) Draw the minimal undirected graph G over which P factors. (This is the undirected graph with the fewest possible edges that contains cliques corresponding to each of the terms in P.)

   (b) For what configuration(s) $x \in \{0, 1\}^6$ is $P(x)$ maximized?

   (c) For what configuration(s) $x \in \{0, 1\}^6$ is $P(x)$ minimized?

2. In a tournament, each player plays three games. Define three random variables:

   - Random variable A is 1 if Tom wins his first game, and is 0 otherwise.
   - Random variable B is 1 if Tom wins his first two games, and is 0 otherwise.
   - Random variable C is 1 if Tom wins all three of his games, and is 0 otherwise.

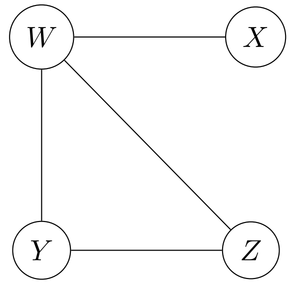
   (a) Is C independent of A (i.e., is it true that for any values a, c that A, C might take on, we have $\Pr(A = a, C = c) = \Pr(A = a)\Pr(C = c)$)? Give a brief (one-sentence) reason for your answer.

   Hint: The answer is no; show this by exhibiting specific choices of a, c for which $\Pr(A = a, C = c) \neq \Pr(A = a)\Pr(C = c)$.

   (b) Is C conditionally independent of A given B (i.e., is it true that $\Pr(C = c|A = a, B = b) = \Pr(C = c|B = b)$)? Give a brief reason.

   (c) Draw an undirected graphical model for (A, B, C).

3. Four random variables W, X, Y, Z factor over the graph shown below.



   (a) What is the functional form of the joint distribution $P(W, X, Y, Z)$?

   (b) Write down a conditional independence property of this distribution.

4. The joint distribution P over (x_1, x_2, x_3) is given by the following table.

x_1	x_2	x_3	Pr
0	0	0	1/3
0	0	1	1/3
1	0	0	1/6
1	1	1	1/6

   (a) One of these variables is a deterministic function of the other two. Identify this relationship.

   (b) Draw the minimal Bayes net G over which this distribution factors. Hint: Start from (a), which shows that one variable is fully determined by the other two; are the latter two variables independent or not?

   (c) Now show how P factors over G by explicitly giving the three (conditional) probability tables, one per x_i.

   (d) Draw the minimal undirected graph G' over which distribution P factors.

5. A Bayesian approach to image denoising. Download the file `gibbs.zip` from the course website. This contains a Jupyter notebook and two images (`noisy_bear.png` and `original_bear.png`). The former is a corrupted version of the latter (which is provided only for reference).

   Let X be the corrupted image and Y the uncorrupted version that we wish to recover. Both are of the form $\{-1, +1\}^{M \times N}$. In the notebook, a Markov random field joint distribution for (X, Y) is given. You will use this to design a Gibbs sampler for $Y|X$.

   (a) Let p denote a specific pixel in the image. Then Y_p is the (reconstructed) value at that pixel, and $Y_{\setminus p}$ is the value at all remaining pixels. Give a precise expression for $\Pr(Y_p = +1 | Y_{\setminus p}, X)$.

   (b) Implement a Gibbs sampler at the appropriate place in the notebook. Initialize Y by setting each pixel to $\{-1, +1\}$ uniformly at random. Then apply the sampler and show the image Y at various stages while the Markov chain is mixing: show any five of the images obtained along the way.

   (c) To reconstruct the image, just set the value of each pixel p to $\mathbb{E}[Y_p | X]$. Show this image.

6. Rejection sampling. Suppose we want to sample from a one-dimensional density $f(x)$ for which we do not have a readymade sampler. One way to do this is via rejection sampling. Here's the idea:

   - We choose another distribution $g(x)$ from which we know how to sample (e.g. uniform or Gaussian distribution). We will call this the proposal distribution. It is important that $g(x)$ be nonzero whenever $f(x)$ is nonzero.
   - Let M be an upper bound on $f(x)/g(x)$; that is, $f(x)/g(x) \leq M$ for all x.
   - To generate a sample from f, we repeat the following procedure until it accepts:
     - Generate a sample x from g
     - With probability $f(x)/(Mg(x))$, accept the sample. Otherwise, reject it.

   Let's see how this works in practice. Let's say we want to sample a number in the range $[0, \pi]$ from

   $$f(x) = \frac{1}{2} \sin x$$

   (think of x as being in radians).

   (a) Plot the density of f.

   (b) Give a suitable proposal distribution $g(x)$. What is a suitable upper bound M on $f(x)/g(x)$?

   (c) Use rejection sampling to generate 10000 samples from f. How many times did you need to sample from g?

   (d) Plot the histogram of the samples you generated, using 50 bins. It should bear a resemblance to the density in (a).